FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2020

TEST DATE: 16-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. It is given that at 20°C, sphere is floating in liquid but at 35°C it begins to sink as on increasing temperature, liquid expands and its density decreases. At 35°C its density becomes equal to that of the sphere and then the buoyancy force on sphere is just equal to the weight of sphere and its is fully submerged in the liquid.

Given that density of sphere is

$$\rho_S = \frac{m}{\frac{4}{3}\pi r^3} = \frac{250}{\frac{4}{3}\times 3.14\times(3.5)^3} = 1.393 \text{ gm/cm}^3$$

It is given that density of liquid at 0°C is ρ_0 = 1.527 gm/cm^3 if γ is the coefficient of cubical expansion of liquid, then its density at 35°C is given as

$$\rho_{35} = \frac{\rho_0}{1+\gamma(35)}$$

At 35°C, as sphere begins to sink, we have density of sphere = density of liquid

$$1.393 = \frac{1.527}{1+\gamma(35)}$$

(As sphere does not expand, its density remains constant)

$$\gamma = \left(\frac{1.527}{1.393}-1\right)\times\frac{1}{35} = 2.75\times10^{-3} \text{ C}^{-1}$$

2. A

Sol. We know that the gas pressure is given as

$$P = \frac{1}{3}\rho v_{rms}^2$$

Now we can write the density of gas as $\rho = m_0N$, where m_0 is the mass of the gas molecule and N is the number of molecules in 1 m^3 of the gas.
In the present case, $N = 100 \times 10^6$

Or $P = \frac{1}{3}(100\times10^6)m_0v_{rms}^2$

As we know rms velocity of gas molecules is given as

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m_0}} \quad \left[\text{As } M = m_0N_{av} \text{ and } \frac{R}{N_{Av}} = k\right]$$

So we have

$P = \frac{1}{3}\times(100\times10^6)\times3kT$ (as T = 3k)

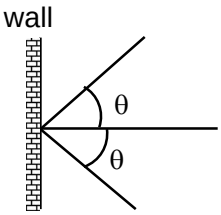
$P = 4.14 \times 10^{-15}$ Pa

3. A

Sol. The induced emf across the inductor is $e = \varepsilon - iR$. Hence the plot of e versus i is a straight line with negative slope.

4. D

Sol. Figure shows a beam of molecules in cylindrical form striking the wall. According to elastic collision, each molecule is reflected from the wall at the same angle of incidence as shown with same speed. If m_0 is the mass of each nitrogen molecule, momentum of each molecule is



$P_0 = m_0v$

During collision, change in momentum of each molecule is

$\Delta P_0 = 2m_0v\cos\theta$

If S is the cross section area of beam and v be the velocity of molecules in the beam then number of molecules incident on wall per second are

$N = n_0vS$ (n_0 = molecule density)

Thus total momentum imparted to the wall per second by the beam and the force exerted on the wall is

$$F = \frac{\Delta P_0}{\Delta t} = 2m_0v\cos\theta\times n_0vS$$

This equation gives total momentum imparted to the wall per second. This is the net fore exerted by beam on the wall.

Thus the pressure on the wall by beam is

$$P = \frac{F\cos\theta}{S} = 2m_0v^2\cos^2\theta\cdot n_0$$

$$= 2\times\frac{28\times10^{-3}}{6.023\times10^{23}}\times9\times10^{24}\times(400)^2\times\cos^2(30^\circ) = 10^5 \text{ Pa}$$

5. A

Sol. Let n be the average number of electrons per unit volume, present in the beam of electrons, then impulse momentum theorem gives

(nAv) (mv) = F

P = F/A

$$n = \frac{P}{(mv^2)}$$

$$\therefore \text{ current } I = nAev = \left(\frac{P}{mv^2}\right)Aev = \frac{PAe}{mv}$$

6. A

Sol. The potential at centre of sphere in which q charge is uniformly distributed through the volume is

$$V_C = \frac{1}{4\pi\varepsilon_0}\frac{3q}{2R}$$

By symmetry, the potential at centre due to half sphere will be half of the complete sphere.

$$V_0 = \frac{V_C}{2} = \frac{1}{4\pi\varepsilon_0}\frac{3q/2}{2R} = \frac{1}{4\pi\varepsilon_0}\frac{3Q}{2R} \text{ (since, } q/2 = Q)$$

7. D

Sol. $V_B - V_A = -\int E_x dx = -$(area under $E_x - x$ curve)

$$V_B - 10 = -\frac{1}{2}\times 2\times(-20) = 20$$

$V_B = 30$ V

8. C

Sol. After the sphere is grounded, potential at centre becomes zero. Let the net charge on the sphere finally be q.

$$\frac{kq}{r} + \frac{k3Q}{x} = 0$$

$$q = -\frac{3Qr}{x}$$

The charge following out of sphere is $\frac{3Qr}{x}$

9. A

Sol. $$R = \int \frac{\rho dx}{A} = \frac{\int_0^{\ell} bxdx}{A} = \frac{b\ell^2}{2A}$$

$$I = \frac{V}{R} = \frac{2VA}{b\ell^2}$$

$$\therefore\ j = \frac{I}{A} = \frac{2V}{b\ell^2}$$

$\Rightarrow j = \sigma E$

$$E = \rho j = \frac{bx2V}{b\ell^2} = \frac{2V}{L^2}x$$

10. A

Sol. With both switches open, $I_A = \frac{E}{R_1 + R_2 + R_3}$

With both switches closed, there will be no current through R_3, the current through E and R_2 is

$$I' = \frac{E}{R_2 + \frac{RR_1}{R + R_1}}$$

Current through the ammeter is

$$I'_A = \frac{RI'}{R + R_1} = \frac{RE}{(R + R_1)R_2 + RR_1}$$

$$= \frac{E}{R_1 + R_2 + \frac{R_2R_1}{R}}$$

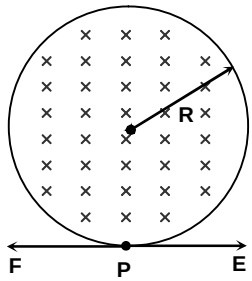
As $I_A = I'_A$

$$R = \frac{R_1R_2}{R_3} = \frac{100 \times 300}{50} = 600\ \Omega$$

11. A

Sol. If we consider the cylindrical surface to be a ring of radius R, there will be an induced electric field due to time varying magnetic field.

$$\int \vec{E} \cdot d\vec{\ell} = -\frac{d\phi}{dt} = \frac{AdB}{dt}$$



$$\Rightarrow E(2\pi R) = A\frac{dB}{dt} = \pi R^2 \frac{dB}{dt}$$

$$E = \frac{R}{2}\frac{dB}{dt}$$

So, force on the electron

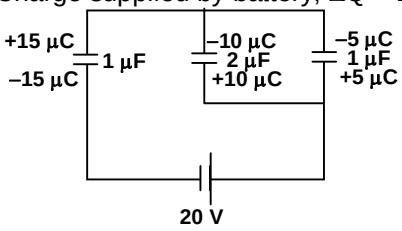
$$F = -Ee = -\frac{eR}{2}\frac{dB}{dt}$$

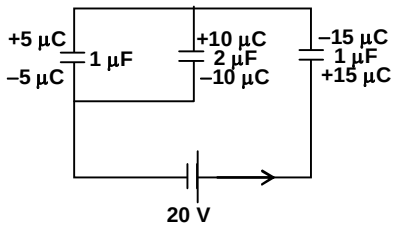
Acceleration = $\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$

Force will be towards left on electron.

12. B

Sol. Charge supplied by battery, $\Delta Q = 10\ \mu C$





Work done by battery

$\Delta W_B = (10\ \mu C)\ (20v) = 200\ \mu J$

Change in potential energy

Stored by capacitors

$\Delta U = U_f - U_i = 0$

Heat loss = $\Delta W_B - \Delta U$

= 200 μJ

13. D

Sol. Magnetic field due to circular segment = $\frac{3}{4}\frac{\mu_0 i}{2a}$

Due to one straight wire segment $= \frac{\mu_0 i}{4\pi b}(\sin 45° + \sin 0°) = \frac{\mu_0 i}{4\sqrt{2}\pi b}$

Net field $= \frac{3\mu_0 i}{8a} + 2 \times \frac{\mu_0 i}{4\sqrt{2}\pi b}$

$= \frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{2a} + \frac{\sqrt{2}}{b}\right)$

14. B

Sol. $q\vec{E} = -q(\vec{v} \times \vec{B})$

$\vec{E} = -(\vec{v} \times \vec{B})$

$\Rightarrow \vec{E} \perp \vec{v}$ and $\vec{E} \perp \vec{B}$

15. B

Sol. $\frac{M}{L} = \frac{Q}{2m}$

$M = \frac{Q}{2m} \times \frac{2}{3} mR^2\omega$

$M = \frac{QR^2\omega}{3}$

16. D

Sol. We take the ampeiran loop of radius r > R.

Since the loop is outside the cylinder, the current through the loop is

$I = \int_0^R kr \times (2\pi r dr) = 2\pi k \int_0^R r^2 dr = \frac{2\pi kR^3}{3}$

$\Rightarrow B \times 2\pi r = \mu_0 I = \frac{2\pi kR^3\mu_0}{3}$

$B = \frac{\mu_0 kR^3}{3r}$

17. C

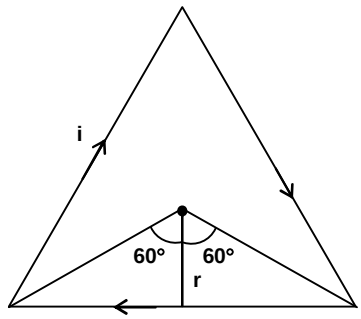
Sol. Due to a wire

$B = \frac{\mu_0}{4\pi} \frac{i}{(\ell / 2\sqrt{3})} (\sin 60° + \sin 60°)$

$= \frac{\mu_0 i}{4\pi\ell} 2\left(\frac{\sqrt{3}}{2}\right) \times 2\sqrt{3} = \frac{6\mu_0 i}{4\pi\ell}$

Magnetic field due to three sides

$B_{net} = 3B = \frac{\mu_0}{4\pi} \frac{18i}{\ell}$



$r = \frac{\ell}{2\sqrt{3}}$

18. C

Sol. $\phi = BS = B\pi R^2$

$= B\pi(R_0 + t)^2$

$e_{ind} = -\frac{d\phi}{dt} = -2\pi(R_0 + t)B$

Since flux is increasing e_{ind} is anticlockwise

19. A

Sol. $\phi = at(T - t) = aTt - at^2$

$e_{ind} = -\frac{d\phi}{dt}$

$e_{ind} = -a(T - 2t)$

$i_{end} = \frac{e_{ind}}{R} = \frac{(2at - aT)}{R}$

$H = \int_0^T i^2 R dt = \int_0^T \frac{(2at - aT)^2}{R} dt$

$= \frac{\int_0^T (4a^2t^2 + a^2T^2 - 4a^2Tt)dt}{R} = \frac{1}{R}\left(\frac{4a^2T^3}{3} + a^2T^3 - \frac{4a^2T^3}{2}\right)$

$H = \frac{a^2T^3}{3R}$

20. A

Sol. $\phi = B\pi r^2 = (16 - 4t^2)\pi r^2$

$e_{ind} = -\frac{d\phi}{dt} = 8\pi r^2 t$

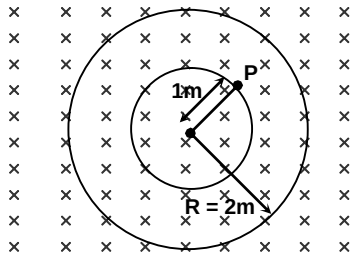
$e_{ind} = \int E \cdot d\ell$

$8\pi r^2 t = E2\pi r$

$\Rightarrow E = \frac{8rt}{2} = 4rt$

at t = 2 sec

E = 8 Volt/m



SECTION – B

21. 2

Sol. $q = \frac{\Delta\phi}{R}$

$\Rightarrow \quad \Delta\phi = qR$ = area of i-t graph $\times$ R

$= \frac{1}{2}(4)(0.1) \times 10 = 0.2 \times 10 = 2$ Weber

22. 5

Sol. $V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$

Here $V = 1.8 \times 10^5$ V, $\theta = 60°$

$r = 50 \times 10^{-2} = 0.5$ m

$\Rightarrow 1.8 \times 10^5 = 9 \times 10^9 \times \frac{p\cos 60°}{(0.5)^2}$

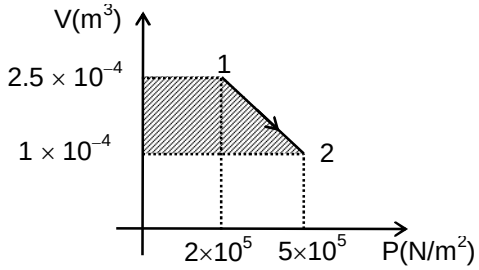
Dipole moment, $p = \frac{1.8 \times 10^5 \times 0.25 \times 2}{9 \times 10^9} = 10^{-5}$ C-m

SECTION – C

23. 00241.50

Sol. We know work done by a gas is given by the area under PV curve or the area between PV-curve and the volume axis. Generally we take in this process volume of gas decreases thus work is done on the gas and it is given as

$$\Delta W = -\frac{1}{2}\times 1.5\times 10^{-4}\times(2+5)\times 10^{5} = -52.5 \text{ Joule}$$



It is given that heat rejected in the process is 70 cal. Thus

$\Delta Q = -70$ cal $= 70 \times 4.2$ J $= -294$ J

Now from first law of thermodynamics, we have

$\Delta Q = \Delta W + \Delta U$

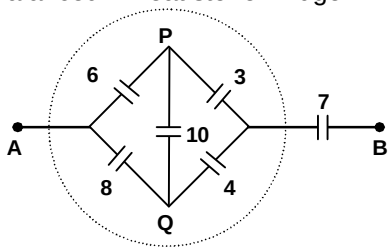
$\Delta U = \Delta Q - \Delta W$

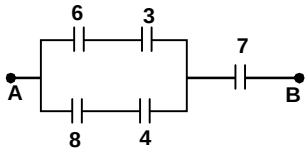
$= (-294) - (-52.5) = -241.5$ J

Thus in the process, internal energy of gas decreases by 241.5 J

24. 00002.80

Sol. Balanced Wheat stone Bridge





$$C_{eq} = \frac{14}{5}\mu F = 2.80\ \mu F$$

25. 10021.34

Sol. We know rms velocity of gas molecules is given as

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

As v_{rms} is directly proportional to $\sqrt{T}$, to double rms speed temperature must be raised to four times, $T_f = 1200$ k. As the gas is enclosed in a container of constant volume, the heat required is

$\Delta Q = \Delta U$

$= nC_V\Delta T$

$= n\frac{5}{2}R\Delta T = \frac{15}{28}\times\frac{5}{2}\times 8.314\times 900$

= 10021.34 Joule

Chemistry

PART – II

SECTION – A

26. C

Sol. $6HCHO + 4NH_3 \longrightarrow (CH_2)_6 N_4$

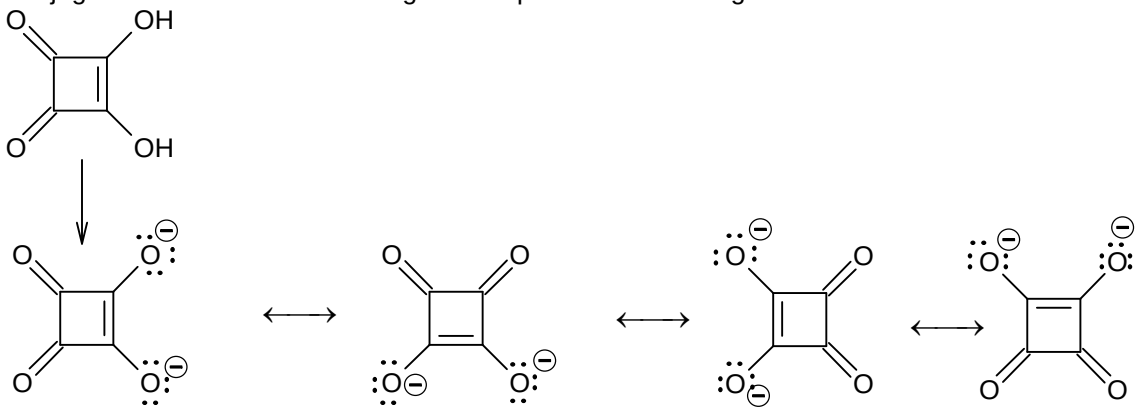
Degree of unsaturation of $(CH_2)_6 N_4 = 3$.

27. D

Sol. Gabriel's phthalimide synthesis involves S_N2 reaction, thus 3° butyl amine cannot be prepared through this method.

28. D

Sol. Conjugate base is stabilized through four equivalent resonating structures.

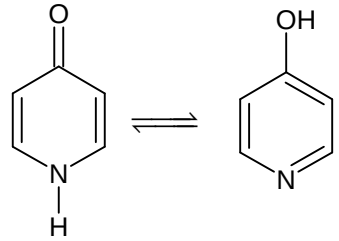


29. C

Sol. As it has no plane of symmetry and point of symmetry.

30. D

Sol.



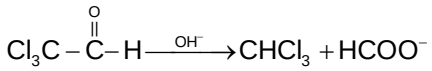
Enol form is aromatic.

31. C

Sol. S_Ni reaction involves retention in configuration.

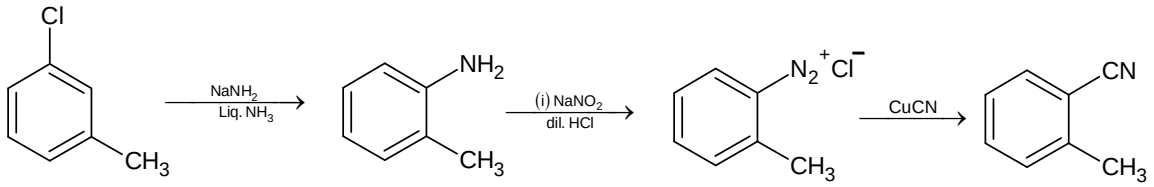
32. C

Sol.



33. B

Sol.



34. D

Sol. As after removal the carbocation becomes aromatic.

35. C

Sol. Glucose can be oxidized by Br_2/H_2O but fructose does not respond in the reaction, as it is a ketose.

36. C

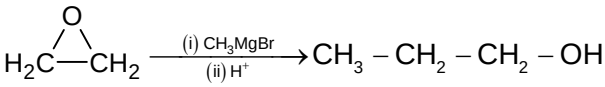
Sol. Due to presence of two amino groups.

37. D

Sol. Extracted of H is the rate determining step in haloform reaction.

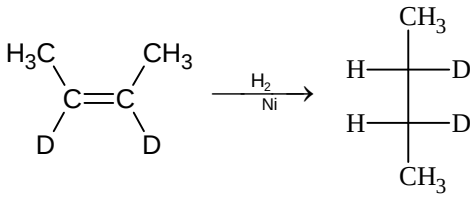
38. C

Sol.



39. D

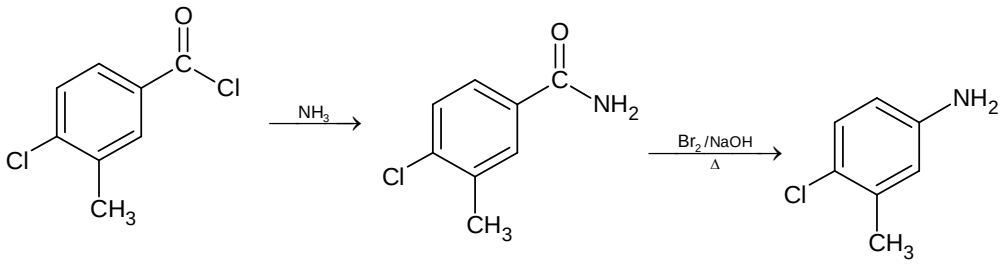
Sol.



If the substrate has 'cis' configuration, on syn addition, a 'meso' compound is generated.

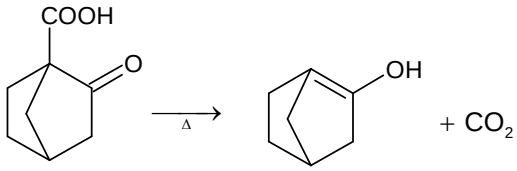
40. B

Sol.



41. C

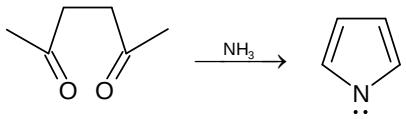
Sol.



After decarboxylation, double bond appears at bridgehead position.

42. D

Sol. $(NH_4)_2CO_3 \longrightarrow 2NH_3 + CO_2 + 2H_2O$



Option (D) is incorrect.

43. D
Sol. Aldol condensation takes place through the formation of enolate anion.

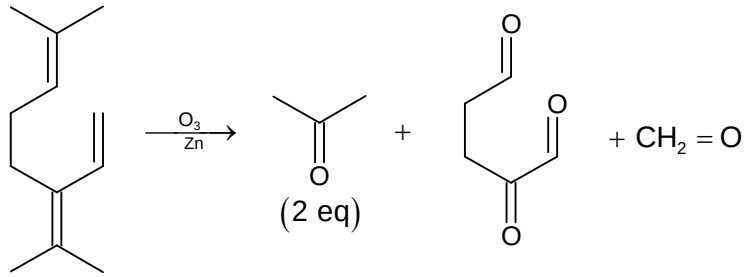
44. C
Sol. A hemiacetal can respond in Fehling's test.

45. D
Sol. A large base always extracts less substituted H.

SECTION – B

46. 3
Sol. All are antiaromatic.

47. 3
Sol.



SECTION – C

48. 00083.00

49. 00099.40

50. 00014.40

Mathematics

PART – III

SECTION – A

51. B
Sol. $\sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) = 0$

52. C
Sol. $\cos(rB - (n - r)A)$ is real part of $e^{i[rB-(n-r)A]}$

53. C
Sol. Centre lies on the line y = x. Let it be (h, h) and radius be r, $r^2 = \frac{4h^2}{\sqrt{4 \cdot 9^2}}$

54. B
Sol. Find foot of perpendicular from (2, 1) on the directrix say (α, β)
Now, image of (α, β) abt tangent is focus
On solving $(\alpha, \beta) = \left(-\frac{1}{2}, -\frac{3}{2}\right)$; focus $\left(\frac{11}{2}, \frac{3}{2}\right)$

55. A
Sol. $x^2 - y^2 = -\cos 2\alpha$ is the director circle which does not exist

56. A
Sol. Let the orthocentre be (h, k), image w.r.t AB will lie on the circle, so $(x - 1)^2 + (y + 1)^2 = 5$

57. D
Sol. $OT^2 = OA \cdot OB = \alpha\beta = \frac{c}{a} \Rightarrow \sqrt{\frac{c}{a}}$

58. A
Sol. $e_1 < 1 < e_2$; f(1) < 0 ; a > 3

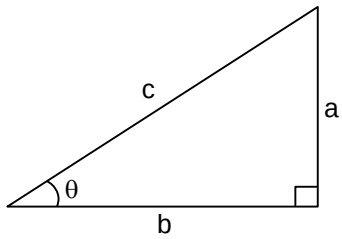
59. B
Sol. Locus is xy = 4

60. A
Sol. Domain [–1, 1] ; f′(x) > 0 ↑
Range is [f(–1), f(1)]

61. D
Sol. $\cos\theta = \frac{x^2 + 8^2 - 15^2}{16x}$, $-1 < \cos\theta < 0$
on solving $7 < x < 13$, $x \in \{8, 9, 10, 11, 12\}$

62. B

Sol. $\frac{1}{a} > \frac{1}{b} > \frac{1}{c}$; $a^2 + b^2 = c^2$, $\frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a^2}$



Solving, we get $1 = \frac{c^2}{a^2} - 2 + \frac{a^2}{c^2}$

Let $t = \frac{a}{c}$; $t^2 + \frac{1}{t^2} - 3 = 0$

$\Rightarrow t = \frac{\sqrt{5}-1}{2} = \frac{a}{c} = \sin\theta$

63. C

Sol. Lines are $y - 1 = \tan\theta(x - 1)$ and $(y - 1) = \cot\theta(x - 1)$ multiply and compare $\sin 2\theta = \frac{2}{a+2}$

64. B

Sol. $y = mx + \frac{4}{m}$ and $hx - ky = h^2 - k^2$ represents same line

65. D

Sol. ${}^nC_2 = {}^nC_3 \Rightarrow n = 5$

66. C

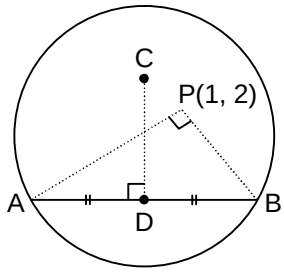
Sol. Equation of asymptotes are $x - 1 = 0$ and $y - 1 = 0$ hyperbola passes through (3, 2)
Equation of hyperbola is $xy = x + y + 1$

67. B

Sol. Curve is hyperbola and $e = \frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$

68. A

Sol. Shown in the figure
AD = BD = PD



69. C

Sol. $\lim_{n\to\infty} \sum_{r=1}^{n} \tan^{-1}(r^2 + r) - \tan^{-1}(r^2 - r) = \tan^{-1}\infty - \tan^{-1}0 = \frac{\pi}{2}$

70. B

Sol. $f(\theta) + f\left(\frac{\pi}{2} - \theta\right) = 1$

SECTION – B

71. 3

Sol. $\cot 120° = \cot (76° + 44°) = -\frac{1}{\sqrt{3}}$

$\cot 76° \cot 44° = 1-\frac{1}{\sqrt{3}}(\cot 76° + \cot 44°)$

72. 1

Sol. $r_1 + r_3 = r_2 - r \Rightarrow \frac{\Delta}{s-a}+\frac{\Delta}{s-c}=\frac{\Delta}{s-b}-\frac{\Delta}{s}$

On solving $\angle B = \frac{\pi}{2}$ and $\angle A + \angle C = 90°$

$\sec^2(90° - C) + \cos^2 90° - \cot^2 C = \operatorname{cosec}^2 C - \cot^2 C = 1$

SECTION – C

73. 00011.00

Sol. $(s - a)(s - b) = 11$ gives $ab = 22 \Rightarrow \Delta = \frac{1}{2}(22) = 11$

74. 00025.00

Sol. Take the image of $(4 + 4\cos\theta, 3 + 3\sin\theta)$ and then eliminate θ

75. 00014.75

Sol. Use parametric form and satisfy in the ellipse, product of roots = $\left|\frac{236}{16+9\sin^2\theta}\right| = \frac{236}{16}$